Optimization of Deep Learning Models to Predict Lung Cancer Indicators Using Chest X-ray Images

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Abstract

Lung cancer is the leading cause of cancer death worldwide, and earlier detection widens the range of effective treatments. Chest radiographs are inexpensive and widely available, but lung lesions on them are easy to miss or confuse with other findings. This study asks how well standard ImageNet-pretrained convolutional neural networks, adapted by transfer learning, can detect the two NIH ChestX-ray14 findings most closely tied to lung tumours: Mass and Nodule. Five backbones (EfficientNetB0, ResNet50, DenseNet201, InceptionV3 and Xception) were fine-tuned at 224×224 pixels with a two-stage schedule and a class-weighted loss, and evaluated on the official NIH test split (25,596 images from 2,797 patients) with patient-level bootstrap confidence intervals. ResNet50 was the strongest single model, with a mean AUROC of 0.788 over 14 findings, a Mass AUROC of 0.771 (95% CI 0.750 to 0.791) and a Nodule AUROC of 0.713 (0.692 to 0.734). An equal-weight ensemble of all five models reached a mean AUROC of 0.793, Mass 0.765 and Nodule 0.716. A validation-only search over 111 ensemble configurations gave no meaningful further gain, and no ensemble was significantly better than ResNet50 on Mass or Nodule. At validation-chosen thresholds, positive predictive value for Mass was only about 0.11 to 0.14, reflecting the low prevalence of the label. The results show that transfer learning gives useful but limited discrimination for these findings and that model averaging mainly helps on the other thoracic findings.

Keywords: lung cancer, chest X-ray, deep learning, transfer learning, ensemble, ChestX-ray14.

1 Introduction

Lung cancer is the foremost cause of cancer-related deaths globally [1]. It arises from uncontrolled growth of abnormal cells in lung tissue, which form tumours that impair normal lung function. Smoking is the primary risk factor, with second-hand smoke, occupational exposures, air pollution and pre-existing lung disease also contributing. Early-stage lung cancer is often asymptomatic, and later symptoms such as persistent cough, chest pain and weight loss resemble those of other illnesses, so the disease frequently goes undetected until an advanced stage. Detection while the disease is still localized allows more treatment options, including surgical resection, and improves prognosis.

The chest X-ray is a fast, non-invasive and low-cost examination, and it is often the first image acquired when lung disease is suspected. Most lung tumours appear on a radiograph as a mass or nodule: a nodule is a small, rounded opacity, and a mass is a larger lesion. Reading these findings is difficult, however. Small nodules are easily hidden behind ribs, the heart or the diaphragm, and many opacities are benign. Automated tools that flag suspicious images could help prioritize radiologist review.

This paper studies how standard deep learning models can be optimized to detect the Mass and Nodule findings on chest radiographs. The contributions are:

- a controlled comparison of five ImageNet-pretrained backbones [2–6], all trained with the same two-stage transfer learning recipe on NIH ChestX-ray14 [7];
- evaluation on the official patient-disjoint NIH test split with patient-level bootstrap confidence intervals and clinically oriented operating points;
- a systematic, validation-only study of ensembling strategies, with paired significance tests against the best single model.

2 Related Work

Wang et al. [7] introduced the ChestX-ray8 database (later expanded to ChestX-ray14) with labels text-mined from radiology reports, and reported baseline weakly supervised classifiers built on ImageNet-pretrained networks. Rajpurkar et al. [8] trained a 121-layer DenseNet (CheXNet) on the 14 findings and reported a mean AUROC of 0.841 and a Mass AUROC of 0.8676; in the same table they quote the Wang et al. baseline at 0.706 for Mass and 0.671 for Nodule. Both studies used random patient-wise splits rather than the official test list, so their numbers are not directly comparable with those reported here.

Several clinical studies have evaluated commercial or research systems for lung lesion detection on radiographs. Yoo et al. [9] found that an AI system improved reader detection of lung cancer on radiographs from the National Lung Screening Trial. Homayounieh et al. [10] reported, in a multicenter reader study, that AI assistance improved human nodule detection accuracy. Li et al. [11] proposed multi-resolution convolutional networks for nodule detection on chest radiographs. These works motivate the use of deep learning as a second reader; the present study focuses on how far standard, openly available backbones and simple ensembles can go on a large public dataset.

3 Data

The study uses the NIH ChestX-ray14 dataset [7]: 112,120 frontal chest radiographs from 30,805 patients. Each image carries zero or more of 14 pathology labels (Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, Pneumothorax) or the label "No Finding". Labels were taken from the April 2020 release of the label table.

The test set is the official NIH test list: 25,596 images from 2,797 patients, containing 1,748 Mass positives (prevalence 6.8%) and 1,623 Nodule positives (6.3%). A validation set of 8,536 images was split from the official train/validation list by patient, so no patient appears in more than one split. All model selection, threshold choice and ensemble weighting used the validation set only; the test set was used once for final reporting.

4 Methods

4.1 Preprocessing

Images were resized to 224×224 pixels and normalized with the preprocessing function matching each backbone's ImageNet weights. Random data augmentation was applied to training images

only. Each image was paired with a 15-dimensional binary target vector (the 14 findings plus No Finding) in a fixed label order that was checked throughout training and evaluation.

4.2 Models

Five convolutional backbones pretrained on ImageNet [12] were compared: ResNet50 [2], DenseNet201 [3], EfficientNetB0 [4], InceptionV3 [5] and Xception [6]. These cover residual connections, dense connectivity, compound scaling, multi-branch inception modules and depthwise separable convolutions. In each model the ImageNet classifier was replaced by a 15-way sigmoid output layer, so that each label is predicted independently (multi-label classification).

4.3 Training

Because positive labels are rare, training used a weighted binary cross-entropy loss with per-label positive and negative weights computed from label frequencies in the training split (with clipping). Training had two stages:

1. **Head training.** The backbone was frozen and only the new output layer was trained for 2 epochs with the Adam optimizer at a learning rate of 10^{-3}.

2. **Fine-tuning.** The top 30% of the backbone layers were unfrozen and the network was trained for 6 further epochs with Adam and a cosine-decay schedule, so the learning rate fell from about 9.3×10^{-5} to 10^{-6}.

After each epoch the mean validation AUROC was computed and the best checkpoint was kept; early stopping on validation AUROC (patience 2) was enabled in stage 2. Each backbone was trained once with a fixed random seed.

4.4 Ensembling

Test and validation predictions of the five models were aligned by image identifier and label name. The primary ensemble is the equal-weight mean of the five models' predicted probabilities. To test whether a more elaborate combination helps, 111 configurations were evaluated over all 31 non-empty subsets of the five models: simple means, weights proportional to validation mean AUROC, a simplex grid search over weights (step 0.1) maximizing validation mean AUROC, per-label rank averaging, and per-label grid weights for Mass and Nodule. All selection used the validation set only.

4.5 Evaluation

The primary metric is the area under the receiver operating characteristic curve (AUROC) for each finding, together with the mean over the 14 findings. 95% confidence intervals were obtained with a nonparametric bootstrap [13] over patients rather than images (1,000 resamples, seed 0), because images from the same patient are correlated. Differences between an ensemble and ResNet50 were assessed with a paired patient-level bootstrap.

For Mass and Nodule, two operating points were set on the validation set and then applied unchanged to the test set: the threshold maximizing Youden's J statistic [14], at which sensitivity, specificity, F1 score and positive predictive value (PPV) are reported; and the threshold giving 90% specificity on validation, at which test sensitivity is reported.

Table 1: Test performance on the official NIH split (25,596 images). AUROC with patient-level bootstrap 95% CIs. Ensemble is the equal-weight mean of all five models.

Model	Mean AUROC (14)	Mass AUROC [95% CI]	Nodule AUROC [95% CI]
ResNet50	0.788	**0.771** [0.750, 0.791]	0.713 [0.692, 0.734]
DenseNet201	0.772	0.735 [0.714, 0.758]	0.698 [0.677, 0.719]
EfficientNetB0	0.771	0.726 [0.703, 0.747]	0.686 [0.664, 0.707]
Xception	0.770	0.743 [0.720, 0.766]	0.696 [0.674, 0.716]
InceptionV3	0.768	0.730 [0.705, 0.752]	0.693 [0.671, 0.715]
Ensemble (mean of 5)	**0.793**	0.765 [0.743, 0.786]	**0.716** [0.695, 0.737]

Table 2: Per-finding test AUROC. Bold marks the best value in each row.

Finding	ResNet50	DenseNet201	EffNetB0	Xception	InceptionV3	Ensemble
Atelectasis	0.749	0.737	0.727	0.732	0.728	**0.751**
Cardiomegaly	0.864	0.830	0.854	0.850	0.842	**0.874**
Consolidation	**0.732**	0.717	0.706	0.721	0.713	0.732
Edema	0.832	0.831	0.832	0.826	0.819	**0.844**
Effusion	0.807	0.792	0.789	0.796	0.788	**0.812**
Emphysema	**0.878**	0.844	0.850	0.852	0.832	0.874
Fibrosis	0.796	0.797	0.786	0.768	0.792	**0.811**
Hernia	0.902	0.897	0.884	0.896	0.907	**0.918**
Infiltration	0.700	0.694	0.699	0.687	0.686	**0.706**
Mass	**0.771**	0.735	0.726	0.743	0.730	0.765
Nodule	0.713	0.698	0.686	0.696	0.693	**0.716**
Pleural Thickening	0.743	0.733	0.733	0.718	0.734	**0.751**
Pneumonia	**0.706**	0.679	0.682	0.664	0.672	0.703
Pneumothorax	0.840	0.829	0.839	0.824	0.812	**0.849**
Mean (14)	0.788	0.772	0.771	0.770	0.768	0.793

5 Results

5.1 Single models

Table 1 summarizes test performance. ResNet50 was the best single model on every summary measure: mean AUROC 0.788, Mass 0.771 and Nodule 0.713. The other four backbones were close to one another (mean AUROC 0.768 to 0.772). Every model detected Mass better than Nodule. Table 2 gives the per-finding AUROCs, and Figure 1 shows the Mass and Nodule ROC curves.

5.2 Operating points

Table 3 reports threshold-based metrics. At the validation Youden threshold, ResNet50 detected 81.6% of Mass cases at 55.2% specificity. Because Mass and Nodule are rare, PPV was low for every model (about 0.11 to 0.14 for Mass, 0.11 to 0.12 for Nodule), meaning most flagged images were false positives. Thresholds set for 90% specificity on validation gave only about 80% to 87% specificity on test, so validation operating points did not transfer exactly.

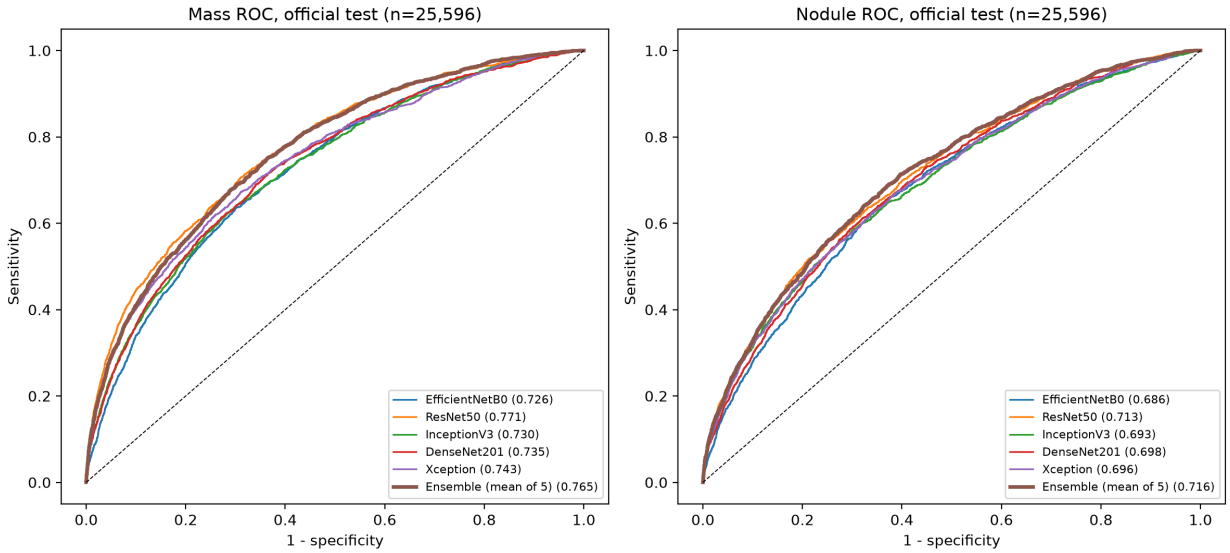


Figure 1: Test ROC curves for Mass and Nodule for the five backbones and the equal-weight ensemble.

Table 3: Mass and Nodule test metrics at validation-chosen thresholds. Sens/Spec/PPV/F1 at the Youden threshold; Sens@90 is test sensitivity at the threshold giving 90% validation specificity.

	Mass					Nodule				
Model	Sens	Spec	PPV	F1	Sens@90	Sens	Spec	PPV	F1	Sens@90
ResNet50	0.816	0.552	0.118	0.206	0.561	0.656	0.641	0.110	0.188	0.386
DenseNet201	0.752	0.586	0.118	0.203	0.504	0.664	0.618	0.105	0.182	0.359
EfficientNetB0	0.755	0.558	0.111	0.194	0.490	0.662	0.619	0.105	0.182	0.344
Xception	0.742	0.604	0.121	0.208	0.518	0.548	0.732	0.121	0.199	0.384
InceptionV3	0.612	0.725	0.140	0.228	0.486	0.612	0.662	0.109	0.185	0.384
Ensemble	0.823	0.541	0.116	0.203	0.562	0.614	0.697	0.121	0.202	0.409

5.3 Ensembles

Table 4 compares the main ensemble variants with ResNet50. The equal-weight mean of all five models raised the mean AUROC from 0.788 to 0.793, and five-model rank averaging gave nearly the same result (0.793, Mass 0.763, Nodule 0.715). The best configuration found by the validation search (grid weights 0.3, 0.2, 0.2, 0.1, 0.2 for ResNet50, EfficientNetB0, Xception, InceptionV3, DenseNet201) reached 0.794, essentially equal to the simple mean. Its gain over ResNet50 on mean AUROC was small but significant (+0.006 [0.004, 0.008]), while its differences on Mass (−0.002 [−0.009, 0.005]) and Nodule (+0.004 [−0.002, 0.010]) were not significant. For the simple mean the corresponding differences were +0.005 [0.003, 0.008] (mean), −0.006 [−0.014, 0.002] (Mass) and +0.003 [−0.004, 0.009] (Nodule). Figure 2 shows ROC curves for the best ensemble.

5.4 Comparison with published work

Table 5 places the results next to two widely cited studies. The numbers are not directly comparable because those studies used random patient-wise splits rather than the official test list. With that

Table 4: Ensemble variants on the test set. Δ columns: paired patient-level bootstrap difference versus ResNet50, with 95% CI.

Configuration	Mean (14) [CI]	Mass [CI]	Nodule [CI]	Δ Mass	Δ Nodule
ResNet50 alone	0.788 [0.781, 0.794]	0.771 [0.750, 0.791]	0.713 [0.692, 0.734]		
Mean of 5	0.793 [0.787, 0.800]	0.765 [0.743, 0.786]	0.716 [0.695, 0.737]	−0.006	+0.003
Rank average of 5	0.793	0.763	0.715		
Grid-weighted (val best)	0.794 [0.787, 0.801]	0.769 [0.748, 0.790]	0.717 [0.696, 0.739]	−0.002	+0.004

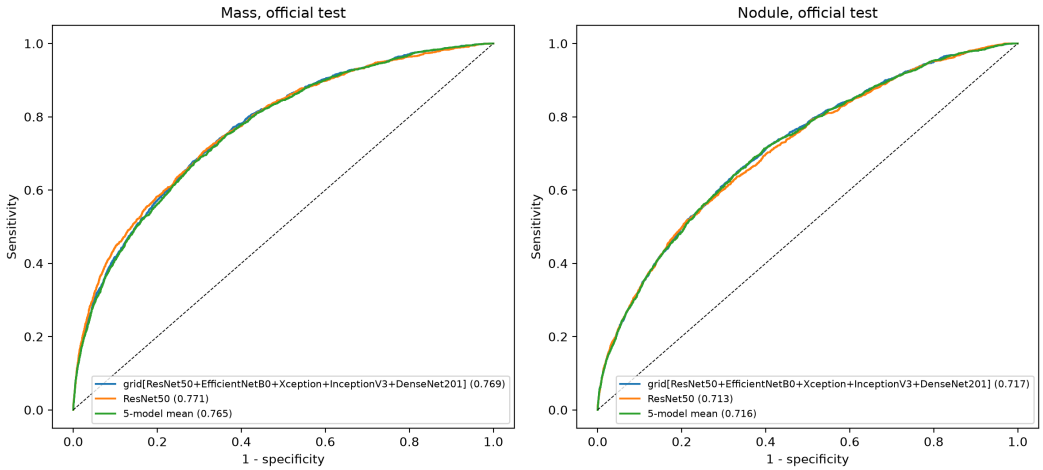


Figure 2: Test ROC curves for Mass and Nodule: validation-selected grid-weighted ensemble, ResNet50 alone and the equal-weight five-model mean.

caveat, the models here exceed the Wang et al. baseline on Mass and Nodule but remain well below CheXNet.

Table 5: Published AUROCs on ChestX-ray14 (different splits; not directly comparable).

Source	Mass	Nodule	Mean (14)
Wang et al. [7], as reported in [8]	0.706	0.671	0.738
CheXNet [8]	0.868	0.780	0.841
This work, ResNet50	0.771	0.713	0.788
This work, mean of 5	0.765	0.716	0.793

6 Discussion

ResNet50 was consistently the strongest backbone, and the five architectures, despite very different designs and sizes, landed within 0.02 of each other in mean AUROC. This suggests that at 224×224 input and a short fine-tuning schedule, data and label quality limit performance more than architecture choice does.

Ensembling helped the average over all 14 findings, but not the findings of main interest. On Mass and Nodule, no ensemble was significantly better than ResNet50 alone; the equal-weight Mass AUROC was even slightly lower. A large search over subsets, weights and rank averaging

added essentially nothing over a plain average. Because 111 configurations were ranked on a single validation set, and the top ten differed by only about 0.001 in validation AUROC, the selected configuration is optimistically biased, and its confidence intervals do not account for the search. For this reason the simple mean is reported as the primary ensemble and ResNet50 as the recommended single model.

Clinically, the low PPV is the most important caveat. With Mass prevalence below 7%, even a reasonably discriminative model flags many more normal images than true positives. Such a model could only be considered as a prioritization aid alongside a radiologist, not as a stand-alone diagnostic test.

Limitations.

- **Label noise.** ChestX-ray14 labels were text-mined from reports and are known to be noisy. Mass and Nodule are radiological findings, not confirmed cancer diagnoses, so they are only a proxy for lung cancer.
- **Resolution.** Images were downsampled to 224×224, which can erase small nodules; this likely explains part of the gap between Mass and Nodule performance.
- **Short training and single runs.** Every model reached its best validation AUROC at its final epoch, which suggests longer training would help. Each backbone was trained once with one seed, so run-to-run variance is not measured.
- **Validation/test shift.** Validation mean AUROC was about 0.024 to 0.033 higher than test for every model (ResNet50: 0.812 vs 0.788), and validation-chosen thresholds did not hold their specificity on test.
- **Ensembles.** The ensemble was not significantly better than ResNet50 on Mass or Nodule, and the weighted search is subject to selection bias.
- **External validity.** All data come from a single hospital system; performance on other populations and scanners is untested.

Future work includes training at higher resolution, longer schedules with multiple seeds, cleaner or radiologist-verified labels, and external validation on independent datasets.

7 Conclusion

Five ImageNet-pretrained backbones were fine-tuned to detect Mass and Nodule, the main lung cancer indicators in ChestX-ray14, and evaluated on the official patient-disjoint test split with patient-level confidence intervals. ResNet50 performed best (Mass AUROC 0.771, Nodule 0.713, mean 0.788). Averaging all five models raised the mean AUROC to 0.793 but did not significantly improve Mass or Nodule detection, and an extensive weighted search added nothing meaningful. Transfer learning with standard backbones gives useful but limited discrimination for these findings, and low positive predictive value means such models are best viewed as triage aids rather than diagnostic tools.

Code Availability

Code for preprocessing, training and evaluation is available at `https://github.com/Aaditya-Shah696/chess-xray-prediction`.

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